Network Toll Solution

TECHNICAL BROCHURE



Representative image: RFID tag customised with Network Toll Solution branding.

RFID TAG FOR WASTE MANAGEMENT

Model: NTS-RT 14 | NFC (13.56 MHz) and UHF (865 - 928 MHz) variants

IP67	Highly Durable	Outdoor Use	UHF / NFC	Water Resistant	Chemical Proof	20 Years Data Retention	Made in India

Description

The NTS-RT 14 is a rugged RFID tag designed for household waste management. It helps track daily waste collection and supports efficient collection operations. Tags are installed outside homes and are read automatically by RFID readers mounted on collection vehicles, or by handheld devices used by waste collectors. Each scan records a collection event and sends the data to municipal or government systems for monitoring and analysis.

The collected data helps authorities optimise collection routes and track waste bin locations. The tag is built for outdoor use and is IP67 rated, ensuring reliable performance in rain, dust, heat and cold. It can be fixed to concrete or metal surfaces using a nail or rivet, and can be customised with a logo, QR code, tag ID and other details.

Key Features

- Proven performance with large-scale deployment across India in waste management initiatives.
- High quality reinforced plastic construction for outdoor conditions, IP67 rated.
- Can be mounted on metal or used off metal; rivet option available.
- Available in NFC (13.56 MHz) or UHF (865 - 928 MHz).
- Fade-free laser printing for tag ID, logo, text and QR code.
- Available in grey, white and black.
- UHF read range: up to 10 m with a fixed reader, up to 7 m with a handheld reader.
- NFC read range: up to 5 cm with any NFC-enabled smartphone.
- Chemical and water resistant.
- Data retention warranty of 20 years (manufacturer warranty).

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Specification

	NFC	UHF
Operating frequency band	NFC 13.56 MHz	UHF 865 MHz or 928 MHz
Chip type	Mifare NFC	Alien Higgs-3
Communication protocol	ISO/IEC 14443-3 (Type A)	ISO/IEC 18000-6C, EPC C1 G2
Memory organisation	1 kB; 16 sectors with 4 blocks per sector; 64 blocks of 16 bytes each	EPC: 96 bits, extendible to 480 bits TID: 96 bits, unalterable and unique User memory: 512 bits Access password: 32 bits Kill password: 32 bits
Read range	Up to 5 cm with NFC-enabled smartphone	Fixed reader: up to **10 m** Handheld reader: up to 7 m

Data retention	20 years
Endurance cycle	100,000 cycles
Size	50 mm x 80 mm x 7 mm
Hole size	Dia 5 mm
Encasement	Engineering plastic
Weight	18 g
Colour	Grey, black, white
Quality testing	100%
Ingress protection	IP67
Operating temperature	-25°C to +85°C
Storage temperature	-25°C to +85°C
Quallication	On and off metal
Attachment / mounting	M5 screw or rivet
Resistance to salt water exposure	Yes
Resistance to lubricant exposure	Yes
Resistance to IPA and thinner	Yes
Encoding	Yes
Logo / text / barcode printing	Yes

Software Solution

Software solution including RFID handheld readers and mobile software for RFID-based waste collection is available through Network Toll Solution.